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Пензенский государственный университет
Кафедра «Вычислительная техника»

Отчет

по лабораторной работе №2
по дисциплине «Программирование на языке Java»
на тему: «Работа с коллекциями объектов»
Вариант 6

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Цель работы

Изучить библиотеку стандартных коллекций Java Collections Framework, позволяющую хранить различные структуры данных.

Лабораторное задание

Модифицировать приложение из предыдущей лабораторной работы, реализовав хранение данных таблицы с использованием библиотеки коллекций. Для этого реализовать класс RecIntegral, способный хранить одну запись таблицы. Для нечетных вариантов в качестве класса-коллекции выбрать ArrayList, для четных - LinkedList. Кроме того, добавить пару кнопок: очистить / заполнить, которые будут очищать таблицу и заполнять ее данными из коллекции соответственно. Оформление лабораторной работы должно быть выполнено в соответствии с требованиями, приведенными в Приложении 2.

Ход работы:

1. Реализовали класс RecIntegral для хранения одной записи таблицы.

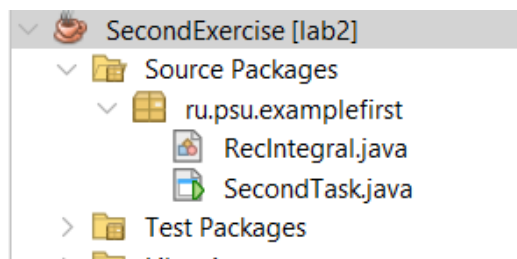


Рисунок 1 - Новый класс RecIntegral

2. Добавили коллекцию LinkedList в программу.

```
LinkedList<RecIntegral> functionList;  
  
public SecondTask() {  
    initComponents();  
    functionList = new LinkedList<>();  
}
```

Рисунок 2 - Добавление коллекции LinkedList

3. Добавили кнопки для очистки таблицы и заполнения таблицы данными из коллекции.

Current function is $\sqrt{x}$

Enter the bottom limit

Add

Clear

Enter the top limit

Delete

Fill

Enter the step

Perform

Bottom limit	Top limit	Step	Result
--------------	-----------	------	--------

Рисунок 3 - Новые кнопки Clear и Fill

Результат работы программы:

Current function is $\sqrt{x}$

Enter the bottom limit

Add

Clear

Enter the top limit

Delete

Fill

Enter the step

Perform

Bottom limit	Top limit	Step	Result
1	2	0.009	
1	8	0.001	
32	40	0.078	
33	114	0.7	

Рисунок 4 - Ввод данных

Current function is $\sqrt{x}$

Enter the bottom limit

Enter the top limit

Enter the step

Add Clear

Delete Fill

Perform

Bottom limit	Top limit	Step	Result
1	2	0.009	1.2062400275427885
1	8	0.001	14.415449632977476
32	40	0.078	47.48244332806577
33	114	0.7	677.6298988109012

Рисунок 5 - Вычисление результатов

Current function is $\sqrt{x}$

Enter the bottom limit

Enter the top limit

Enter the step

Add Clear

Delete Fill

Perform

Bottom limit	Top limit	Step	Result
1	2	0.009	1.2062400275427885
1	8	0.001	14.415449632977476
32	40	0.078	47.48244332806577

Рисунок 6 - Удаление последней строки из таблицы и коллекции

Current function is $\sqrt{x}$

Enter the bottom limit

Enter the top limit

Enter the step

Add Clear

Delete Fill

Perform

Bottom limit	Top limit	Step	Result
--------------	-----------	------	--------

Рисунок 7 - Очистка таблицы

Current function is $\sqrt{x}$

Enter the bottom limit

Enter the top limit

Enter the step

Add Clear

Delete Fill

Perform

Bottom limit	Top limit	Step	Result
1.0	2.0	0.009	1.2062400275427885
1.0	8.0	0.001	14.415449632977476
32.0	40.0	0.078	47.48244332806577

Рисунок 8 - Заполнение таблицы из коллекции

Вывод

В ходе выполнения данной работы мы изучили библиотеку стандартных коллекций Java Collections Framework, позволяющую хранить различные структуры данных, а также поработали с коллекциями ArrayList и LinkedList.

Листинг

Файл SecondTask.java

```
/*  
  
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to  
change this license  
  
 * Click nbfs://nbhost/SystemFileSystem/Templates/GUIForms/JFrame.java to edit  
this template  
  
*/  
  
package ru.psu.examplefirst;  
  
  
import java.util.LinkedList;  
  
import javax.swing.JOptionPane;  
  
import javax.swing.table.DefaultTableModel;  
  
  
/**  
  
 *  
  
 * @author Asus  
  
 */
```

```
public class SecondTask extends javax.swing.JFrame {
```

```
/**
```

```
 * Creates new form FirstExample
```

```
 */
```

```
LinkedList<RecIntegral> functionList;
```

```
public SecondTask() {
```

```
    initComponents();
```

```
    functionList = new LinkedList<>();
```

```
}
```

```
/**
```

```
 * This method is called from within the constructor to initialize the form.
```

```
 * WARNING: Do NOT modify this code. The content of this method is always
```

```
 * regenerated by the Form Editor.
```

```
 */
```

```
@SuppressWarnings("unchecked")
```

```
// <editor-fold defaultstate="collapsed" desc="Generated Code">//GEN-BEGIN: initComponents
```

```
private void initComponents() {
```

```
jPanel = new javax.swing.JPanel();

jScrollPane1 = new javax.swing.JScrollPane();

jTable = new javax.swing.JTable();

txtBot = new javax.swing.JTextField();

txtTop = new javax.swing.JTextField();

txtStep = new javax.swing.JTextField();

labelBot = new javax.swing.JLabel();

labelTop = new javax.swing.JLabel();

labelStep = new javax.swing.JLabel();

btnAdd = new javax.swing.JButton();

btnDelete = new javax.swing.JButton();

btnPerform = new javax.swing.JButton();

labelFunction = new javax.swing.JLabel();

labelF = new javax.swing.JLabel();

btnClear = new javax.swing.JButton();

btnFill = new javax.swing.JButton();
```

```
setDefaultCloseOperation(javax.swing.WindowConstants.EXIT_ON_CLOSE);
```

```
jTable.setModel(new javax.swing.table.DefaultTableModel(
    new Object [][] {
```



```
    },  
  
    new String [] {  
        "Bottom limit", "Top limit", "Step", "Result"  
    }  
)  
{  
    boolean[] canEdit = new boolean [] {  
        true, true, true, false  
    };  
  
    public boolean isCellEditable(int rowIndex, int columnIndex) {  
        return canEdit [columnIndex];  
    }  
});  
  
jScrollPane1.setViewportView(jTable);  
  
labelBot.setText("Enter the bottom limit");  
  
labelTop.setText("Enter the top limit");  
  
labelStep.setText("Enter the step");  
  
btnAdd.setText("Add");
```

```
btnAdd.addActionListener(new java.awt.event.ActionListener() {  
    public void actionPerformed(java.awt.event.ActionEvent evt) {  
        btnAddActionPerformed(evt);  
    }  
});
```

```
btnDelete.setText("Delete");  
  
btnDelete.addActionListener(new java.awt.event.ActionListener() {  
    public void actionPerformed(java.awt.event.ActionEvent evt) {  
        btnDeleteActionPerformed(evt);  
    }  
});
```

```
btnPerform.setText("Perform");  
  
btnPerform.addActionListener(new java.awt.event.ActionListener() {  
    public void actionPerformed(java.awt.event.ActionEvent evt) {  
        btnPerformActionPerformed(evt);  
    }  
});
```

```
labelFunction.setFont(new java.awt.Font("Segoe UI", 2, 12)); // NOI18N  
labelFunction.setText("Current function is");
```

```
labelF.setFont(new java.awt.Font("Segoe UI", 3, 18)); // NOI18N
```

```
labelF.setText(" √x");
```

```
btnClear.setText("Clear");
```

```
btnClear.addActionListener(new java.awt.event.ActionListener() {
```

```
    public void actionPerformed(java.awt.event.ActionEvent evt) {
```

```
        btnClearActionPerformed(evt);
```

```
    }
```

```
});
```

```
btnFill.setText("Fill");
```

```
btnFill.addActionListener(new java.awt.event.ActionListener() {
```

```
    public void actionPerformed(java.awt.event.ActionEvent evt) {
```

```
        btnFillActionPerformed(evt);
```

```
    }
```

```
});
```

```
    javax.swing.GroupLayout jPanelLayout = new  
    javax.swing.GroupLayout(jPanel);
```

```
    jPanel.setLayout(jPanelLayout);
```

```
    jPanelLayout.setHorizontalGroup(
```

```
jPanelLayout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
```

```
    .addGroup(jPanelLayout.createSequentialGroup())
```

```
        .addContainerGap()
```

```
        .addComponent(jScrollPane1)
```

```
        .addContainerGap()
```

```
    .addGroup(jPanelLayout.createSequentialGroup())
```

```
        .addGap(67, 67, 67)
```

```
.addGroup(jPanelLayout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
```

```
    .addGroup(jPanelLayout.createSequentialGroup())
```

```
.addGroup(jPanelLayout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
```

```
    .addComponent(labelStep,  
    javax.swing.GroupLayout.Alignment.TRAILING)
```

```
    .addComponent(labelBot,  
    javax.swing.GroupLayout.Alignment.TRAILING)
```

```
    .addComponent(labelTop,  
    javax.swing.GroupLayout.Alignment.TRAILING))
```

```
    .addGap(18, 18, 18)
```

```
.addGroup(jPanelLayout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
```

```
    .addComponent(txtBot,  
        javax.swing.GroupLayout.PREFERRED_SIZE, 130,  
        javax.swing.GroupLayout.PREFERRED_SIZE)
```

```
    .addComponent(txtTop,  
        javax.swing.GroupLayout.PREFERRED_SIZE, 130,  
        javax.swing.GroupLayout.PREFERRED_SIZE)
```

```
    .addComponent(txtStep,  
        javax.swing.GroupLayout.PREFERRED_SIZE, 130,  
        javax.swing.GroupLayout.PREFERRED_SIZE)))
```

```
    .addGroup(jPanelLayout.createSequentialGroup()
```

```
        .addComponent(labelFunction)
```

```
    .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
```

```
        .addComponent(labelF,  
            javax.swing.GroupLayout.PREFERRED_SIZE, 37,  
            javax.swing.GroupLayout.PREFERRED_SIZE)))
```

```
    .addGap(135, 135, 135)
```

```
.addGroup(jPanelLayout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING, false)
```

```
    .addComponent(btnAdd, javax.swing.GroupLayout.DEFAULT_SIZE,  
        javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
```

```

        .addComponent(btnDelete,
javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)

        .addComponent(btnPerform,
javax.swing.GroupLayout.DEFAULT_SIZE, 100, Short.MAX_VALUE))

.addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.UNRELATED)

.addGroup(jPanelLayout.createParallelGroup(javax.swing.GroupLayout.Alignmen
t.LEADING, false)

        .addComponent(btnClear,
javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)

        .addComponent(btnFill, javax.swing.GroupLayout.DEFAULT_SIZE,
100, Short.MAX_VALUE))

.addContainerGap(20, Short.MAX_VALUE))

);

jPanelLayout.setVerticalGroup(

jPanelLayout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)

        .addGroup(javax.swing.GroupLayout.Alignment.TRAILING,
jPanelLayout.createSequentialGroup()

        .addGap(8, 8, 8)

```

```
.addGroup(jPanelLayout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
```

```
    .addComponent(labelFunction)
```

```
    .addComponent(labelF))
```

```
.addGap(19, 19, 19)
```

```
.addGroup(jPanelLayout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
```

```
    .addComponent(txtBot,  
    javax.swing.GroupLayout.PREFERRED_SIZE,  
    javax.swing.GroupLayout.DEFAULT_SIZE,  
    javax.swing.GroupLayout.PREFERRED_SIZE)
```

```
    .addComponent(labelBot)
```

```
    .addComponent(btnAdd)
```

```
    .addComponent(btnClear))
```

```
.addGap(27, 27, 27)
```

```
.addGroup(jPanelLayout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
```

```
    .addComponent(txtTop,  
    javax.swing.GroupLayout.PREFERRED_SIZE,  
    javax.swing.GroupLayout.DEFAULT_SIZE,  
    javax.swing.GroupLayout.PREFERRED_SIZE)
```

```
    .addComponent(labelTop)
```

```
    .addComponent(btnDelete)
```

```

        .addComponent(btnFill))

    .addGap(27, 27, 27)

    .addGroup(jPanelLayout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE))

        .addComponent(txtStep,
            javax.swing.GroupLayout.PREFERRED_SIZE,
            javax.swing.GroupLayout.DEFAULT_SIZE,
            javax.swing.GroupLayout.PREFERRED_SIZE)

        .addComponent(labelStep)

        .addComponent(btnPerform))

    .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED, 49,
        Short.MAX_VALUE)

        .addComponent(jScrollPane1,
            javax.swing.GroupLayout.PREFERRED_SIZE, 220,
            javax.swing.GroupLayout.PREFERRED_SIZE)

        .addContainerGap())

    );

    javax.swing.GroupLayout layout = new
    javax.swing.GroupLayout(getContentPane());

    getContentPane().setLayout(layout);

    layout.setHorizontalGroup(

```



```

layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)

    .addComponent(jPanel, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE)

);

layout.setVerticalGroup(

```

```

layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)

    .addComponent(jPanel, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE)

);

```

```

pack();

} // </editor-fold> // GEN-END: initComponents

```

```

private void btnAddActionPerformed(java.awt.event.ActionEvent evt) { // GEN-
FIRST:event_btnAddActionPerformed

```

```

// TODO add your handling code here:

```

```

DefaultTableModel model = (DefaultTableModel)jTable.getModel();

model.addRow(new Object[]{txtBot.getText(), txtTop.getText(),
txtStep.getText()});

```

```

functionList.add(new RecIntegral(

```

```
        Double.parseDouble(txtBot.getText()),  
        Double.parseDouble(txtTop.getText()),  
        Double.parseDouble(txtStep.getText())  
    ));
```

```
txtBot.setText("");  
txtTop.setText("");  
txtStep.setText("");
```

```
}//GEN-LAST:event_btnAddActionPerformed
```

```
private void btnDeleteActionPerformed(java.awt.event.ActionEvent evt)  
{//GEN-FIRST:event_btnDeleteActionPerformed
```

```
    // TODO add your handling code here:
```

```
    DefaultTableModel model = (DefaultTableModel)jTable.getModel();
```

```
    try {
```

```
        int selectedRowID = jTable.getSelectedRow();
```

```
        model.removeRow(selectedRowID);
```

```
        functionList.remove(selectedRowID);
```

```
    } catch (Exception e){
```

```
        JOptionPane.showMessageDialog(null, e);
```

```
    }
```

```
}//GEN-LAST:event_btnDeleteActionPerformed
```

```

private void btnPerformActionPerformed(java.awt.event.ActionEvent evt)
{
    //GEN-FIRST:event_btnPerformActionPerformed

        // TODO add your handling code here:

        DefaultTableModel model = (DefaultTableModel)jTable.getModel();

        //    double botLimit, toplimit, step;

        double result;

        int selectedRowID = jTable.getSelectedRow();


        //    botLimit = Double.parseDouble(jTable.getValueAt(selectedRowID,
        //0).toString());

        //    toplimit = Double.parseDouble(jTable.getValueAt(selectedRowID,
        //1).toString());

        //    step = Double.parseDouble(jTable.getValueAt(selectedRowID,
        //2).toString());


        RecIntegral integral = functionList.get(selectedRowID);

        result = integral.Integral();

        model.setValueAt(result, selectedRowID, 3);

    }
    //GEN-LAST:event_btnPerformActionPerformed

```

```
private void btnClearActionPerformed(java.awt.event.ActionEvent evt) {//GEN-FIRST:event_btnClearActionPerformed
```

```
    // TODO add your handling code here:
```

```
    DefaultTableModel model = (DefaultTableModel)jTable.getModel();
```

```
    model.setRowCount(0);
```

```
}//GEN-LAST:event_btnClearActionPerformed
```

```
private void btnFillActionPerformed(java.awt.event.ActionEvent evt) {//GEN-FIRST:event_btnFillActionPerformed
```

```
    // TODO add your handling code here:
```

```
    DefaultTableModel model = (DefaultTableModel)jTable.getModel();
```

```
    Object[] data = new Object[4];
```

```
    for (int i = 0; i < functionList.size(); i++){
```

```
        data[0] = functionList.get(i).getBotLimit();
```

```
        data[1] = functionList.get(i).getTopLimit();
```

```
        data[2] = functionList.get(i).getStep();
```

```
        data[3] = functionList.get(i).getResult();
```

```
        model.addRow(data);
```

```
    }
```

```
}//GEN-LAST:event_btnFillActionPerformed
```

```
/**
```

```

* @param args the command line arguments

*/

public static void main(String args[]) {

    /* Set the Nimbus look and feel */

    //<editor-fold defaultstate="collapsed" desc=" Look and feel setting code
(optional) ">

    /* If Nimbus (introduced in Java SE 6) is not available, stay with the default
look and feel.

    * For details see
http://download.oracle.com/javase/tutorial/uiswing/lookandfeel/plaf.html

    */

    try {

        for (javax.swing.UIManager.LookAndFeelInfo info :
javax.swing.UIManager.getInstalledLookAndFeels()) {

            if ("Nimbus".equals(info.getName())) {

                javax.swing.UIManager.setLookAndFeel(info.getClassName());

                break;

            }

        }

    } catch (ClassNotFoundException ex) {

        java.util.logging.Logger.getLogger(SecondTask.class.getName()).log(java.util.log
ging.Level.SEVERE, null, ex);

    } catch (InstantiationException ex) {

```

```
java.util.logging.Logger.getLogger(SecondTask.class.getName()).log(java.util.logging.Level.SEVERE, null, ex);
```

```
    } catch (IllegalAccessException ex) {
```

```
java.util.logging.Logger.getLogger(SecondTask.class.getName()).log(java.util.logging.Level.SEVERE, null, ex);
```

```
    } catch (javax.swing.UnsupportedLookAndFeelException ex) {
```

```
java.util.logging.Logger.getLogger(SecondTask.class.getName()).log(java.util.logging.Level.SEVERE, null, ex);
```

```
    }
```

```
//</editor-fold>
```

```
//</editor-fold>
```

```
/* Create and display the form */
```

```
java.awt.EventQueue.invokeLater(new Runnable() {
```

```
    public void run() {
```

```
        new SecondTask().setVisible(true);
```

```
    }
```

```
});
```

```
}
```

```
// Variables declaration - do not modify//GEN-BEGIN:variables

private javax.swing.JButton btnAdd;

private javax.swing.JButton btnClear;

private javax.swing.JButton btnDelete;

private javax.swing.JButton btnFill;

private javax.swing.JButton btnPerform;

private javax.swing.JPanel jPanel;

private javax.swing.JScrollPane jScrollPane1;

private javax.swing.JTable jTable;

private javax.swing.JLabel labelBot;

private javax.swing.JLabel labelF;

private javax.swing.JLabel labelFunction;

private javax.swing.JLabel labelStep;

private javax.swing.JLabel labelTop;

private javax.swing.JTextField txtBot;

private javax.swing.JTextField txtStep;

private javax.swing.JTextField txtTop;

// End of variables declaration//GEN-END:variables

}
```

Файл RecIntegral.java

```
/*
```

```
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to  
change this license
```

```
 * Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Class.java to edit this  
template
```

```
*/
```

```
package ru.psu.examplefirst;
```

```
/**
```

```
 *
```

```
 * @author Asus
```

```
*/
```

```
public class RecIntegral {
```

```
    private double botLimit, topLimit, step, result;
```

```
    public RecIntegral(double botLimit, double topLimit, double step){
```

```
        this.botLimit = botLimit;
```

```
        this.topLimit = topLimit;
```

```
        this.step = step;
```

```
        this.result = 0.0;
```

```
    }
```



```

private static double Function(double x){

    return java.lang.Math.sqrt(x);

}


public double Integral(){

    int n = (int)((topLimit - botLimit) / step);

    for (int i = 0; i < n - 1; i++){

        double x0 = botLimit + i * step;

        double x1 = botLimit + (i + 1) * step;

        result += (Function(x0) + Function(x1)) * step / 2;

    }


    double last = botLimit + n * step;

    if (last < topLimit){

        double lastTrapezoid = (Function(last) + Function(topLimit)) * (topLimit -
last) / 2;

        result += lastTrapezoid;

    }


    return result;

}

```

```
public void setBotLimit(double botLimit){  
    this.botLimit = botLimit;  
}
```

```
public double getBotLimit(){  
    return botLimit;  
}
```

```
public void setTopLimit(double topLimit){  
    this.topLimit = topLimit;  
}
```

```
public double getTopLimit(){  
    return topLimit;  
}
```

```
public void setStep(double step){  
    this.step = step;  
}
```

```
public double getStep(){  
    return step;  
}
```

```
}
```

```
public double getResult(){
```

```
    return result;
```

```
}
```

```
}
```